Microbe isolation: environmental sample → primary plate → pure culture → preservation vials

Guan Jie Phang, Yin-Tse Huang · HuangLab (NCKU)

(v2.0, 2026-10-20; data steps via /lims-register (Stage 1 → 2a → 2b → Culture#))

Source: rewritten from the lab's protocols.io protocol *Step by Step from Environmental Samples to Preservation Vials* (dx.doi.org/10.17504/protocols.io.n2bvj8mk5gk5/v1). Bench steps unchanged; the Specify datasheet upload is replaced by `/lims-register`. Do not modify the bench steps without telling the lab manager.

Route: environmental sample (Stage 1, Specimen#) → primary plate (Stage 2a) → one secondary plate per isolate (Stage 2b, Culture#) → preservation vials + a lysis tube for DNA extraction (Stage 3a).

Materials :

1. Media plates: MEA for primary plates, PDA for secondary plates (see *Fungal culture media*)
2. Scalpel, tweezers, spreader; 95% ethanol cooling bottle; alcohol lamp; 75% ethanol sprayer
3. Grafting tape; PP bags no. 8 / 10 / 12 (8 / 24–32 / 48–56 plates)
4. Dissecting microscope; flatbed scanner (EPSON Scan) and the PlateSegmentation PC
5. Preservation vials: water vials ×2 and glycerol vials ×2 per Culture#; 2 mL lysis tube with beads (see *DNA extraction*)
6. Black marker or technical pen; freezing boxes (water vials: steel cabinet by Prof. Huang's office; glycerol vials: −80 °C freezer outside N927)

Procedure

Stage 1 · register the specimens

---

1. Catalogue every container that arrived as **one Stage-1 specimen** in LIMS: run `/lims-register` with your sample list (spreadsheet, CSV or a photo of the labelled bags/tubes). Method = `Soil`, `Fungal strain`, etc.; collector, date, locality as known. Confirm the preview; LIMS mints the **Specimen#** and prints the labels.

※ **Note:** A strain collection (many isolates from one source) is registered as **one umbrella Stage-1 specimen**; do not split by collection event.

Stage 2a · isolate primary cultures

---

2. Sterilise the isolation area (air and bench) with the 75% ethanol sprayer; make sure a mist is emitted. Move the equipment back in afterwards.

---

3. Before plating, register the **primary plate** in LIMS: `/lims-register` Stage 2a (one plate per umbrella specimen; media `MEA`, method `Direct isolation`, number of plates). Write the plate ID, media, date and your name on the **bottom** of each plate.

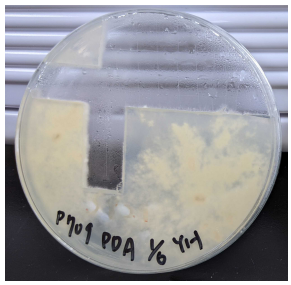


Plate number, media, date, your name

---

4. Sterilise every tool before each isolation: dip the scalpel / tweezers / spreader in 95% ethanol and flame; **repeat three times**.

---

5. Isolate microbes from the sample onto the primary plates. The method depends on the sample type (direct plating, dilution, baiting): check the literature or ask a senior.

---

6. Seal each plate with two rounds of grafting tape, bag them (PP bag no. 8 / 10 / 12) and incubate in your incubator slot for **7–21 days**.



---

7. Check growth daily; **do not let plates overgrow**. Subculture as soon as colonies are mature.

Stage 2b · pick and subculture isolates

---

8. Under the dissecting microscope choose the target colonies. Mark each on the plate bottom with its future **secondary plate number**. Two morphotypes on one primary plate = **two** secondary numbers.

---

9. Register the isolates: `/lims-register` Stage 2b, **one row per isolate** (source = the Stage-2a plate; media `PDA`; strain code → `morph_description`; provenance → notes). Review the preview, then confirm: LIMS mints one **Culture#** per isolate, in row order, and creates the water-vial preps. Registered rows cannot be deleted, so check the list first.

---

10. Subculture each marked colony (single colony or hyphal tip) onto a labelled PDA plate following steps 3–7. These are the **secondary cultures**. Keep the primary plates until every secondary plate is clean; if a secondary plate is contaminated, redo it from the primary.

Plate images

---

11. Sterilise the scanner glass with 75% ethanol, then wipe with detergent water (prevents condensation). Photograph the plate order (groups of **6**), remove lids, place plates **upside down** on the glass, keep the order.

---

12. EPSON Scan → Professional mode → File Save Settings: **.jpg**. Scan. Move the files to `D:\PlateSegmentation\images`.

---

13. Anaconda Navigator → environment `onxx` → Open Terminal: `D:` → `cd PlateSegmentation` → `python ONXX_predict.py`. Output in `D:\PlateSegmentation\output`. Rename each image with its plate number and upload to **HuangLab Collection / Plate picture / 20XX**; attach the file name to the Culture# in LIMS.

Preservation (Stage 2b → vials)

---

14. After **7–21 days** the secondary cultures are ready. Per **Culture#** prepare **2 water vials + 2 glycerol vials** and **one 2 mL lysis tube** with beads. Label every cap with the Culture# (black marker / technical pen).

---

15. UV the laminar flow for **15 min**; ethanol-sterilise everything going in; light the alcohol lamp. Tools as for isolation, but a **scalpel** instead of tweezers and spreader.

---

16. Flame the scalpel before every cut. Slice agar from one colony into **one water vial + one glycerol vial**; the remaining agar goes into the lysis tube for DNA extraction (Stage 3a, *DNA extraction v9.0*).

---

17. Store: water vials straight into their freezing box (steel cabinet). Glycerol vials in a freezing container at **−20 °C for 1 day**, then into the **−80 °C** collection (outside N927). Record vial positions in LIMS.

※ **Note:** The lysis tube is the input for Stage 3a: register the extraction as a draft, quantify on the Qubit, and only then mint the MolWork#.